DESCRIPTION OF COURSEWORK

Course Code	EEE411
Course Name	Communication System
Lecturer	Dr. Khuzairi Bin Masrakin
Academic Session	2024/09
Assessment Title	Performance Analysis of Digital Modulation

A. Introduction/ Situation/ Background Information

Digital modulation encodes a digital information signal into the transmitted signal's amplitude, phase, or frequency. The encoding process affects the bandwidth of the transmitted signal and its robustness to channel impairments. In general, a modulation technique encodes several bits into one symbol, and the symbol transmission rate determines the transmitted signal's bandwidth. Since the signal bandwidth is determined by the symbol rate, having a large number of bits per symbol generally yields a higher data rate for a given signal bandwidth. However, the larger the number of bits per symbol, the greater the required received Signal-to-Noise Ratio (SNR) for a given target Bit Error Rate (BER).

Digital modulation techniques may be linear or nonlinear. In linear modulation, the transmitted signal’s amplitude and/or phase varies linearly with the digital modulating signal, whereas the transmitted signal amplitude is constant for nonlinear techniques. Linear modulation techniques, including all forms of Quadrature Amplitude Modulation (QAM) and Phase Shift Keying (PSK), use less bandwidth than nonlinear techniques, including various forms of frequency/minimum-shift-keying (FSK and MSK).

GNU Octave is a high-level language, primarily intended for numerical computations. It provides a convenient command-line interface for solving linear and nonlinear problems numerically, and for performing other numerical experiments using a language that is mostly compatible with Matlab. GNU Octave is also freely redistributable software. GNU Octave may be redistributed and/or modified under the terms of the GNU General Public License (GPL) as published bythe Free Software Foundation. The latest version of GNU Octave is available for download at https://www.gnu.org/software/octave/download.

B. Course Learning Outcomes (CLO) covered

The assignment aims to study the baseband digital modulated signal's properties and investigate the channel noise's effect on the Bit Error Rate (BER).

At the end of this assessment, students are able to:

CLO 2 Analyse major communication subsystems and investigate the performances of various communication systems

C. University Policy on Academic Misconduct

1. Academic misconduct is a serious offense in Xiamen University Malaysia. It can be defined as any of the following:
   i. **Plagiarism** is submitting or presenting someone else's work, words, ideas, data or information as your own intentionally or unintentionally. This includes incorporating published and unpublished material, whether in manuscript, printed or electronic form into your work without acknowledging the source (the person and the work).
   ii. **Collusion** is two or more people collaborating on a piece of work (in part or whole) which is intended to be wholly individual and passed it off as own individual work.
   iii. **Cheating** is an act of dishonesty or fraud in order to gain an unfair advantage in an assessment. This includes using or attempting to use, or assisting another to use materials that are prohibited or inappropriate, commissioning work from a third party, falsifying data, or breaching any examination rules.

2. All assessments submitted must be the student's own work, without any materials generated by AI tools, including direct copying and pasting of text or paraphrasing. Any form of academic misconduct, including using prohibited materials or inappropriate assistance, is a serious offense and will result in a zero mark for the entire assessment or part of it. If there is more than one guilty party, such as in case of collusion, all parties involved will receive the same penalty.

D. Instruction to Students

1. This is a group assignment. You are required to work in group of two to four members.
2. There are **FOUR** tasks in this assignment. Attempt ALL.
3. You are required to use GNU Octave / Matlab to complete this assignment.
4. Deadline: **22 December 2024, 11:59:59 pm**
5. Use the cover page provided.
6. Submit the softcopy of your report to Dr. Khuzairi Bin Masrakin (khuzairi.masrakin@xmu.edu.my)

Note:

* If you are using GNU Octave, the exercises will need to use the GNU Octave communication package. The package must be manually loaded into the environment by typing the command "pkg load communications" command in the Command Window. The command only has to be executed once every time the GNU Octave is loaded into the operating system.

E. Evaluation Breakdown

No.	Component Title	Percentage (%)
1.	GNU Octave / Matlab scripts	20
2.	Project report	80
	TOTAL	**100**

F. Task(s)

Task 1: Generating digital data and symbols

The script below generates a stream of random binary data and then groups the bit stream into symbols.

```
clear all
clc

%% Section 1
k = 2;
mlevel = 2^k;
N = k*1000;

%% Section 2
x = randi([0 1], N, 1);
x_index = (1:N)';
```

```
%% Section 3
xsym = bi2de(reshape(x,k,length(x)/k).','left-msb');
b = -2.*mod(xsym,k)+k-1;
a = 2.*floor(xsym./k)-k+1;
xmod = a + 1i.*b;

%% Section 4
Tx_x = xmod;
SNR = 18;
Tx_awgn = awgn(Tx_x, SNR, 'measured');

%% Section 5
figure(1)
clf
nsym = 50;
subplot(2,6,[1:4]);
stem(x_index(1:nsym*k), x(1:nsym*k), 'filled');
title('Plot #1: Signal Bit Stream');
xlabel('Bit Index');
ylabel('Binary Value');
axis([0 nsym*k 0 max(x)*1.1])
set(gca,'ytick',0:1)

subplot(2,6,[7:10]);
stem(x_index(1:nsym), xsym(1:nsym), 'filled');
title('Plot #2: Signal Symbol Stream');
xlabel('Symbol Index');
ylabel('Integer Value');
axis([0 nsym 0 max(xsym)*1.1])
set(gca,'ytick',0:round(max(xsym)))

subplot(2,6,5:6);
plot(real(xmod),imag(xmod),'ro','MarkerFaceColor','r', 'MarkerSize', 5);
axis([-max(real(xmod))*1.25 max(real(xmod))*1.25 -max(imag(xmod))*1.25
max(imag(xmod))*1.25]);
title('Plot #3: Constellation diagram (Without AWGN)');
grid on

subplot(2,6,11:12);
plot(real(Tx_awgn),imag(Tx_awgn),'bo','MarkerFaceColor','b', 'MarkerSize',
5);
axis([-max(real(Tx_awgn))*1.25 max(real(Tx_awgn))*1.25 -
max(imag(Tx_awgn))*1.25 max(imag(Tx_awgn))*1.25]);
title('Plot #4: Constellation diagram (With AWGN)');
grid on
```

Run the script above in GNU Octave / Matlab and answer the following questions.

Exercises

1. Discuss what the code block in Sections 1 – 3 does.
2. Interpret the SNR in Section 4.

3. Discuss what you understand about the plots (Plot #1 – #5) plotted in Section 5.
4. For 4-QAM, analyze the changes in the constellation diagram of the QAM signal with AWGN for SNR values of 50 dB, 18 dB, 10 dB, and 5 dB. Conclude your findings.
5. Modify the script to generate 16-QAM symbols and observe the changes in the plot. Discuss your findings.
6. Compare the constellation diagram with AWGN of 4-QAM and 16-QAM with SNR of 18 dB and 10 dB. Discuss your findings.

** Include the snapshot of all figures in your report.*

Task 2: Generating digital modulated baseband waveform

The script below generates the baseband waveform of digitally modulated symbols.

```
clear all
clc

%% Section 1
k = 2;
mlevel = 2^k;
N = k*256;
rate = 270.8333;                    % symbol rate
fs = 1e5;                           % sampling frequency
ts = 1/fs;                          % sampling interval
npersymbol = round(1/rate*fs);      % sample points per symbol

%% Section 2
x = randi([0 1], N, 1);

%% Section 3
xsym = bi2de(reshape(x,k,length(x)/k).','left-msb');

%% Section 4
xp = linspace(0,N/k-1,N/k);
xf = linspace(0,N/k-1,N/k*npersymbol);
b = -2.*mod(xsym,k)+k-1;
a = 2.*floor(xsym./k)-k+1;
b_interp = interp1(xp, b, xf, "spline");
a_interp = interp1(xp, a, xf, "spline");
xmod_interp = a_interp + i.*b_interp;
n_interp = N/k*npersymbol;    % number of sample after interpolation
t = linspace(0,(n_interp-1)*ts,n_interp);

%% Section 5
Tx_x = xmod_interp;
SNR = 50;
Tx_awgn = awgn(Tx_x, SNR, 'measured');

%% Section 6
freq = linspace(-fs/2, fs/2, n_interp);
p = fftshift(fft(Tx_awgn));
p = (1/(fs*n_interp)) * abs(p).^2;
p = 20*log10(p) + 30;
p = p - max(p);

%% Section 7
figure(1)
clf
subplot(2,6,[1:4]);
plot(t*1e3, real(Tx_awgn), t*1e3, imag(Tx_awgn));
title('Plot #1: Modulated waveform (Time Domain)');
xlabel('Time, ms');
ylabel('Amplitude');
axis([0 max(t)*1e3 -max(real(Tx_awgn))*1.1 max(real(Tx_awgn))*1.1])
grid on

%% Section 8
```

```
subplot(2,6,5:6);
plot(real(Tx_awgn),imag(Tx_awgn),'go','MarkerFaceColor',[0,1,0],
'MarkerSize', 0.1);
axis([-max(real(Tx_awgn))*1.25 max(real(Tx_awgn))*1.25 -
max(imag(Tx_awgn))*1.25 max(imag(Tx_awgn))*1.25]);
title('Plot #2: Constellation diagram');
grid on

subplot(2,1,2)
plot(freq, p)
title('Plot #3: Spectrum');
xlabel('Frequency, Hz');
ylabel('Normalized Power Level');
axis([-rate*1.5 rate*1.5 -120 10])
grid on
```

Run the script above in GNU Octave / Matlab and answer the following questions.

Exercises

1. Discuss what the code block does in Section 4. Why it is needed?
2. Discuss what you understand about the graphs (Plot #1, Plot #2, and Plot #3).
3. What are the differences between the constellation plot (Plot #2) in this task and the one (Plot #4) in Task 1?
4. Estimate the bandwidth of the signal. Evaluate the effect of different symbol rates and the number of bits per symbol on the signal's bandwidth.

* *Include the snapshot of all figures in your report.*

Task 3: Demodulating the signal

The script below has added the demodulation of the digitally modulated signal.

```
clear all
clc

%% Section 1
k = 2;
mlevel = 2^k;
N = k*2048;

%% Section 2
x = randi([0 1], N, 1);
x_index = 0:N;

%% Section 3
xsym = bi2de(reshape(x,k,length(x)/k).','left-msb');
xmod = qammod(xsym,mlevel);

%% Section 4
Tx_x = xmod;
SNR = 18;
Tx_awgn = awgn(Tx_x, SNR, 'measured');

%% Section 5
Rx_x = Tx_awgn;
Rx_x_demod = qamdemod(Rx_x,mlevel);

figure(1)
clf

%% Plot Tx & Rx symbol
nsym = 100;          % number of symbols to plot
subplot(1,7,1:4);
stem(x_index(1:nsym), xsym(1:nsym), 'r', 'filled', 'MarkerSize', 10);
hold on
stem(x_index(1:nsym), Rx_x_demod(1:nsym), 'filled', 'b', 'MarkerSize', 5);
hold off
title('Plot #1: Transmitted and Received Symbol');
xlabel('Symbol Index');
ylabel('Level');
axis([0 nsym 0 max(xsym)*1.1])
set(gca,'ytick',0:round(max(xsym)))
legend('Tx', 'Rx');

%% Plot Tx & Rx constellation diagram
subplot(1,7,5:7);
plot(real(Rx_x),imag(Rx_x), 'bo', 'MarkerFaceColor', 'b', 'MarkerSize', 5);
hold on
plot(real(Tx_x),imag(Tx_x), 'ro', 'MarkerFaceColor', 'r', 'MarkerSize', 5);
hold off
axis([-max(real(Rx_x))*1.25 max(real(Rx_x))*1.25 -max(imag(Rx_x))*1.25
max(imag(Rx_x))*1.25]);
title('Plot #2: Constellation Diagram');
grid on
legend('Rx', 'Tx');
```

Run the script above in GNU Octave and answer the following questions.

Exercises

1. Discuss what you understand about the plots (Plot #1 and Plot #2).
2. For 4-QAM, analyze the changes in the constellation diagram of the QAM signal with AWGN for SNR values of 50 dB, 18 dB, 10 dB, and 5 dB. Compare the transmitted and received symbols for each SNR value. Conclude your findings.
3. Repeat the same analysis for 16-QAM, 64-QAM, and 256-QAM. Conclude your findings.

** Include the snapshot of all figures in your report.*

Task 4: BER estimation

The script below calculates the Bit Error Rate (BER).

```
clear all
clc

%% Section 1
k = 2;
mlevel = 2^k;
N = k*2048;

%% Section 2
x = randi([0 1], N, 1);
x_index = 0:N;

%% Section 3
xsym = bi2de(reshape(x,k,length(x)/k).');
xmod = qammod(xsym,mlevel);

%% Section 4
Tx_x = xmod;
SNR = 25;
Tx_awgn = awgn(Tx_x, SNR, 'measured');

%% Section 5
Rx_x = Tx_awgn;
Rx_x_demod = qamdemod(Rx_x,mlevel);

%% Section 6
z = de2bi(Rx_x_demod);
Rx_x_BitStream = reshape(z.',numel(z),1);

%% Calculate and print BER (in Command Window)
[number_of_errors, bit_error_rate] = biterr(x, Rx_x_BitStream);
disp(['Number of error bit = ' num2str(number_of_errors)])
disp(['Bit error rate = ' num2str(bit_error_rate*100) '%'])

figure(1)
clf

%% Plot Tx & Rx symbol
nsym = 100;          % number of symbols to plot
subplot(2,7,[1:5]);
stem(x_index(1:nsym), xsym(1:nsym), 'r', 'filled', 'MarkerSize', 10);
hold on
stem(x_index(1:nsym), Rx_x_demod(1:nsym), 'filled', 'b', 'MarkerSize', 5);
hold off
title('Plot #1: Transmitted and Received Symbol');
xlabel('Symbol Index');
ylabel('Level');
axis([0 nsym 0 max(xsym)*1.1])
set(gca,'ytick',0:round(max(xsym)))
legend('Tx', 'Rx');

%% Plot Tx & Rx bit stream
subplot(2,7,8:12);
```

```
stem(x(1:nsym*k), 'r', 'filled', 'MarkerSize', 10);
hold on
stem(Rx_x_BitStream(1:nsym*k), 'filled', 'b', 'MarkerSize', 5);
hold off
title('Plot #2: Transmitted and Received Bit Stream');
xlabel('Symbol Index');
ylabel('Bit');
axis([0 nsym 0 max(Rx_x_BitStream)*1.1])
set(gca,'ytick',0:round(max(xsym)))
legend('Tx', 'Rx');

%% Plot Tx constellation diagram
subplot(2,7,6:7);
plot(real(Tx_x),imag(Tx_x),'ro', 'MarkerFaceColor','r', 'MarkerSize', 5);
axis([-max(real(Tx_x))*1.25 max(real(Tx_x))*1.25 -max(imag(Tx_x))*1.25
max(imag(Tx_x))*1.25]);
title('Plot #3: Tx Constellation Diagram');
grid on

%% Plot Rx constellation diagram
subplot(2,7,13:14);
plot(real(Rx_x),imag(Rx_x),'bo', 'MarkerFaceColor','b', 'MarkerSize', 5);
axis([-max(real(Rx_x))*1.25 max(real(Rx_x))*1.25 -max(imag(Rx_x))*1.25
max(imag(Rx_x))*1.25]);
title('Plot #4: Rx Constellation Diagram');
grid on
```

Run the script above in GNU Octave and answer the following questions.

Exercises

1. Study the relationship between Bit Error Rate (BER) over SNR and QAM levels.
2. Determine how much SNR is required for 4-QAM, 16-QAM, 64-QAM, and 256 QAM to get a BER of less than 1%.

* *Include the snapshot of all figures in your report.*

Report Format:

1. Introduction
2. Objective(s)
3. Results for Task 1, Task 2, Task 3, and Task 4 (Please include all results such as, but not limited to, the codes (if there are any changes), the graphs generated, and the answers to all the exercises.
4. Discussion on what you have learned (learning outcomes)
5. Conclusion

Warning: The scripts generate random binary data. The data pattern is different every time you run it. Results of the same binary pattern with others will be treated as plagiarism.

APPENDIX 1

MARKING RUBRICS

Component Title	Group report				**Percentage (%)**		100	
Criteria	**Score and Descriptors**						**Weight (%)**	**Marks**
	Excellent (5)	**Good (4)**	**Average (3)**	**Fair (2)**	**Poor (1)**	**Very Bad (0)**		
Introduction	Well-written	Satisfactory	Need minority improvement	Need major revision	Poorly written	Not found in the report	5	
Objective(s)	Well-defined	Satisfactory	Need minority improvement	Need major revision	Poorly defined	Not found in the report	5	
Task 1 Results/figures	Well-presented	Satisfactory	Need minority improvement	Need major revision	Wrong results	Not found in the report	5	
Task 1 Discussion	Well written	Satisfactory	Need minority improvement	Need major revision	Poorly written/wrong	Not found in the report	15	
Task 2 Results/figures	Well-presented	Satisfactory	Need minority improvement	Need major revision	Wrong results	Not found in the report	5	
Task 2 Discussion	Well written	Satisfactory	Need minority improvement	Need major revision	Poorly written/wrong	Not found in the report	15	
Task 3 Results/figures	Well-presented	Satisfactory	Need minority improvement	Need major revision	Wrong results	Not found in the report	5	
Task 3 Discussion	Well written	Satisfactory	Need minority improvement	Need major revision	Poorly written/wrong	Not found in the report	15	
Task 4 Results/figures	Well-presented	Satisfactory	Need minority improvement	Need major revision	Wrong results	Not found in the report	5	
Task 4 Discussion	Well written	Satisfactory	Need minority improvement	Need major revision	Poorly written/wrong	Not found in the report	15	
Learning outcome	Well written	Satisfactory	Need minority improvement	Need major revision	Poorly written	Not found in the report	5	
Conclusion	Well written	Satisfactory	Need minority improvement	Need major revision	Poorly written	Not found in the report	5	
						TOTAL	**100**	

APPENDIX 2

Description of Complex Problem Solving (CPS)

The assignment is a complex engineering problem. Its mapping to the CLO and PLO are as follows.

1) Assignment 2 → CLO2 → PLO2

The complex problem-solving attributes involved in the assignment are as follows.

Attribute	Remarks
WP1 Depth of knowledge required	The assignment is a complex engineering problem that cannot be resolved without in-depth knowledge of electrical communication system.
WP3 Depth of analysis required	The assignment is a problem-based design scenario. There is no obvious solution and requires abstract thinking. Each group has to simulate the performance of different modulation schemes (GNU Octave/Matlab) and do some analysis.